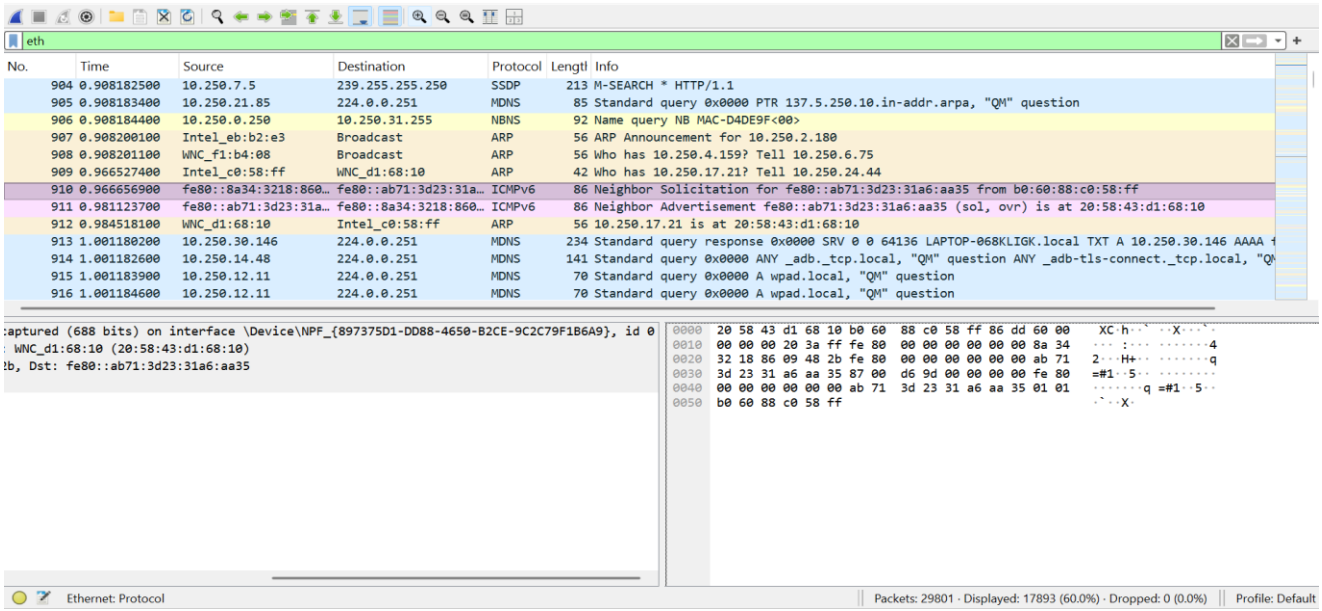
Task 2: Ethernet frame inspection

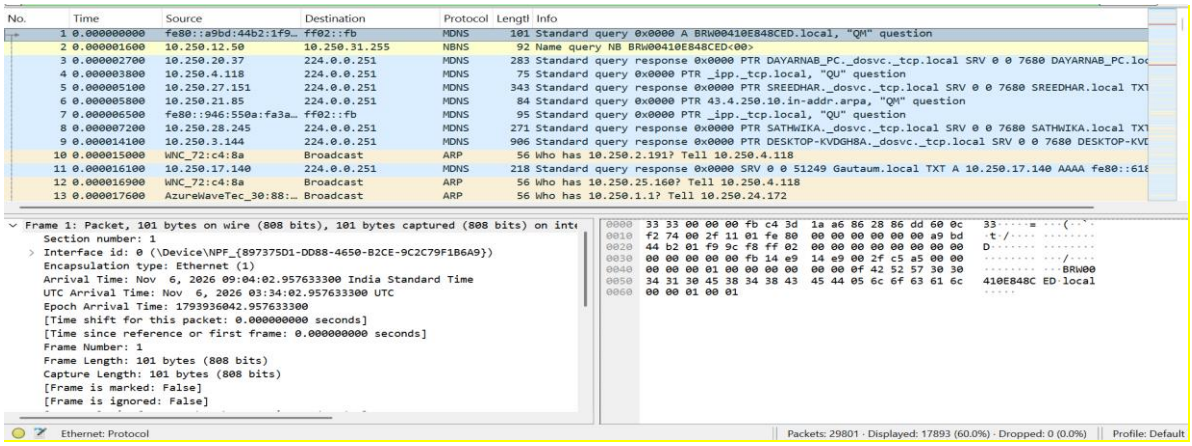
1. filter eth



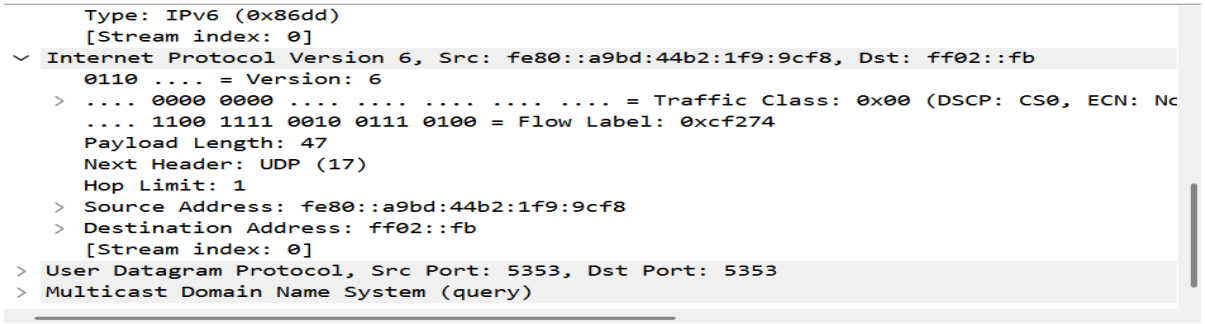
2. five different packets

Packet 1

Frame number & frame length

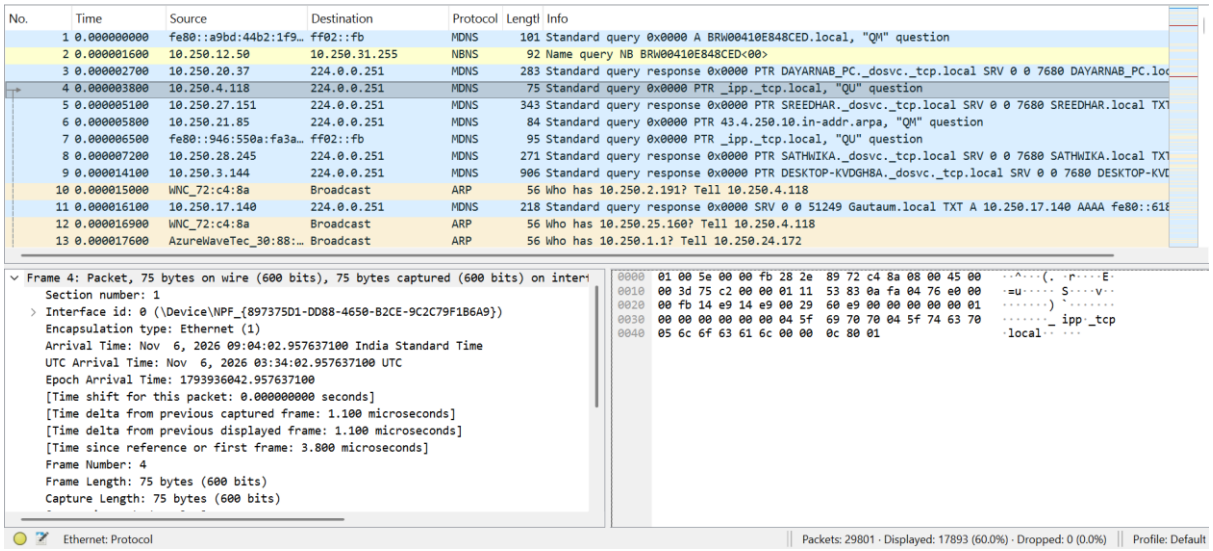


Ether Type, Source & destination address

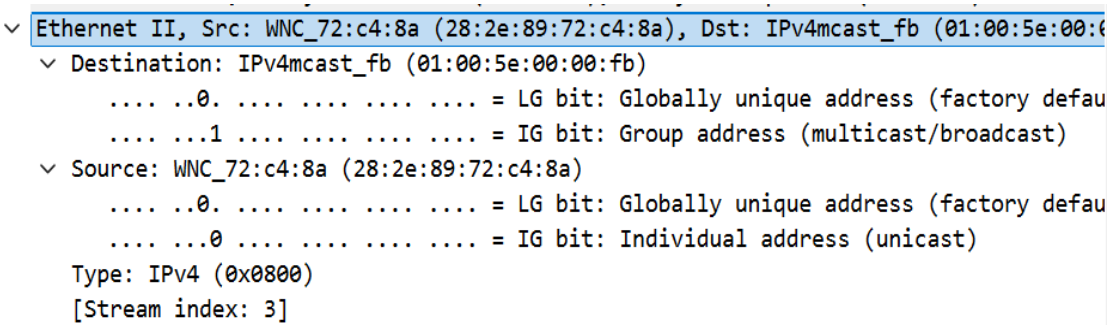


Packet 2

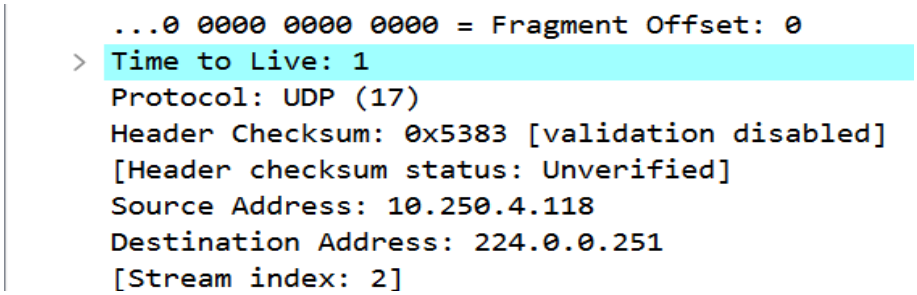
Frame number & length



Ether Type

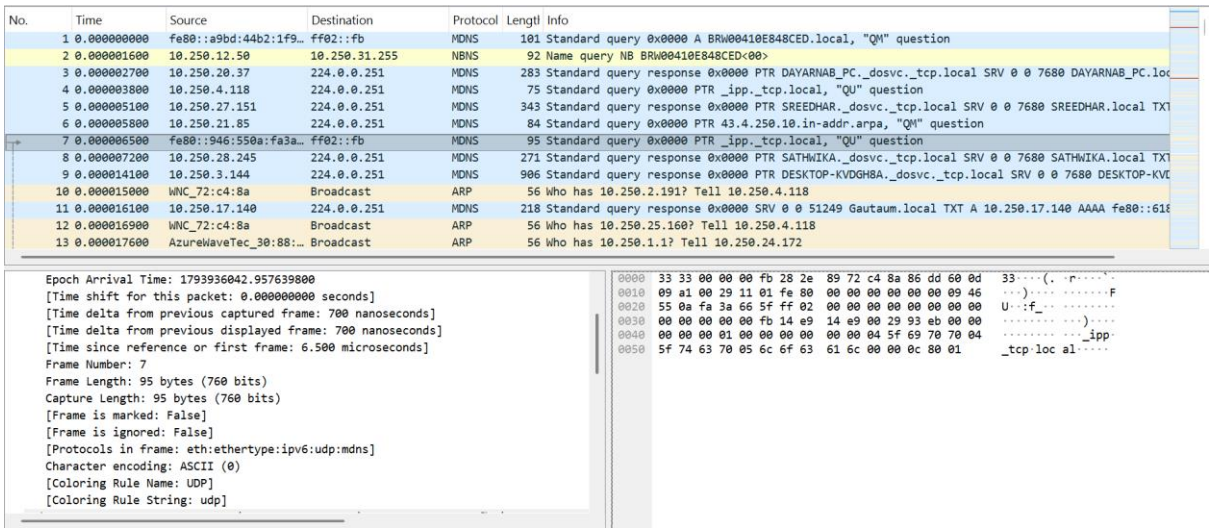


Source & destination address



Packet 3

Frame number & length



No.	Time	Source	Destination	Protocol	Lengtl	Info
1	0.000000000	fe80::a9bd:44b2:1f9…	ff02::fb	MDNS	101	Standard query 0x0000 A BRW00410E848CED.local, "QM" question
2	0.000001600	10.250.12.50	10.250.31.255	NBNS	92	Name query NB BRW00410E848CED<00>
3	0.000002700	10.250.20.37	224.0.0.251	MDNS	283	Standard query response 0x0000 PTR DAYARNAB_PC._dosvc._tcp.local SRV 0 0 7680 DAYARNAB_PC.loc
4	0.000003800	10.250.4.118	224.0.0.251	MDNS	75	Standard query 0x0000 PTR _ipp._tcp.local, "QU" question
5	0.000005100	10.250.27.151	224.0.0.251	MDNS	343	Standard query response 0x0000 PTR SREEDHAR._dosvc._tcp.local SRV 0 0 7680 SREEDHAR.local TXT
6	0.000005800	10.250.21.85	224.0.0.251	MDNS	84	Standard query 0x0000 PTR 43.4.250.10.in-addr.arpa, "QM" question
7	0.000006500	fe80::946:550a:fa3a…	ff02::fb	MDNS	95	Standard query 0x0000 PTR _ipp._tcp.local, "QU" question
8	0.000007200	10.250.28.245	224.0.0.251	MDNS	271	Standard query response 0x0000 PTR SATHWIKA._dosvc._tcp.local SRV 0 0 7680 SATHWIKA.local TXT
9	0.000014100	10.250.3.144	224.0.0.251	MDNS	906	Standard query response 0x0000 PTR DESKTOP-KVDGH8A._dosvc._tcp.local SRV 0 0 7680 DESKTOP-KVD
10	0.000015000	WNC_72:c4:8a	Broadcast	ARP	56	Who has 10.250.2.191? Tell 10.250.4.118
11	0.000016100	10.250.17.140	224.0.0.251	MDNS	218	Standard query response 0x0000 SRV 0 0 51249 Gautaum.local TXT A 10.250.17.140 AAAA fe80::618
12	0.000016900	WNC_72:c4:8a	Broadcast	ARP	56	Who has 10.250.25.160? Tell 10.250.4.118
13	0.000017600	AzureWaveTec_30:88:…	Broadcast	ARP	56	Who has 10.250.1.1? Tell 10.250.24.172

```
Epoch Arrival Time: 1793936042.957639800
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 700 nanoseconds]
[Time delta from previous displayed frame: 700 nanoseconds]
[Time since reference or first frame: 6.500 microseconds]
Frame Number: 7
Frame Length: 95 bytes (760 bits)
Capture Length: 95 bytes (760 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:udp:mdns]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
```

```
0000  33 33 00 00 00 fb 28 2e  89 72 c4 8a 86 dd 60 0d   33····(. ·r····`·
0010  09 a1 00 29 11 01 fe 80  00 00 00 00 00 00 09 46   ···)···· ·······F
0020  55 0a fa 3a 66 5f ff 02  00 00 00 00 00 00 00 00   U··:f_·· ········
0030  00 00 00 00 00 fb 14 e9  14 e9 00 29 93 eb 00 00   ········ ···)····
0040  00 00 00 01 00 00 00 00  00 00 04 5f 69 70 70 04   ········ ···_ipp·
0050  5f 74 63 70 05 6c 6f 63  61 6c 00 00 0c 80 01      _tcp·loc al·····
```

Ether Type

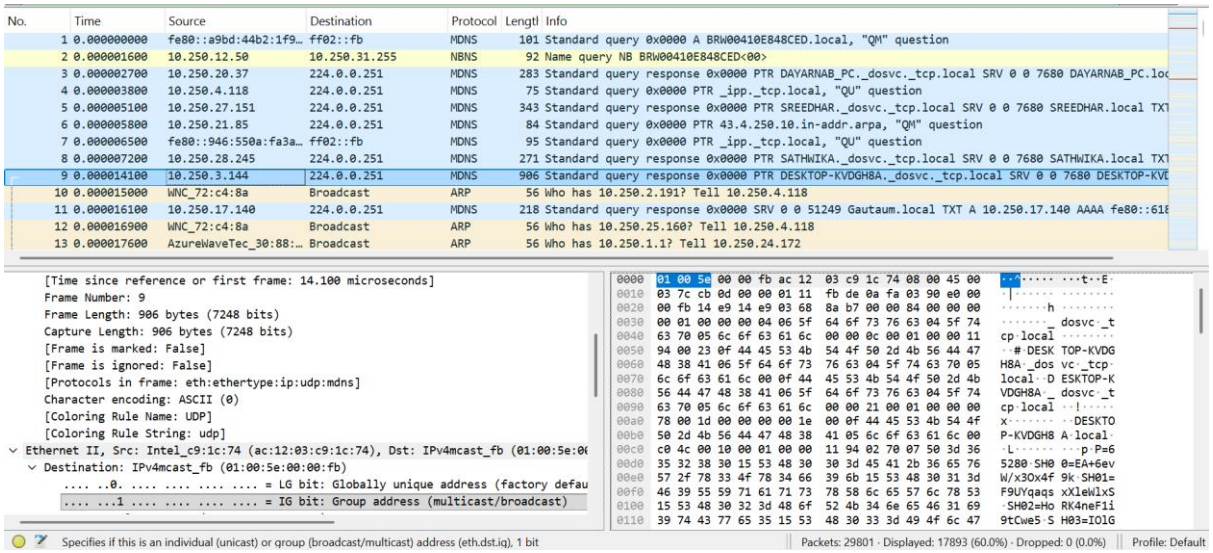
```
▾ Ethernet II, Src: WNC_72:c4:8a (28:2e:89:72:c4:8a), Dst: IPv6mcast_fb (33:33:00:00:(
  ▾ Destination: IPv6mcast_fb (33:33:00:00:00:fb)
      .... ..1. .... .... .... .... = LG bit: Locally administered address (this is
      .... ...1 .... .... .... .... = IG bit: Group address (multicast/broadcast)
  ▾ Source: WNC_72:c4:8a (28:2e:89:72:c4:8a)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory defau
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv6 (0x86dd)
    [Stream index: 6]
```

Source & destination Address

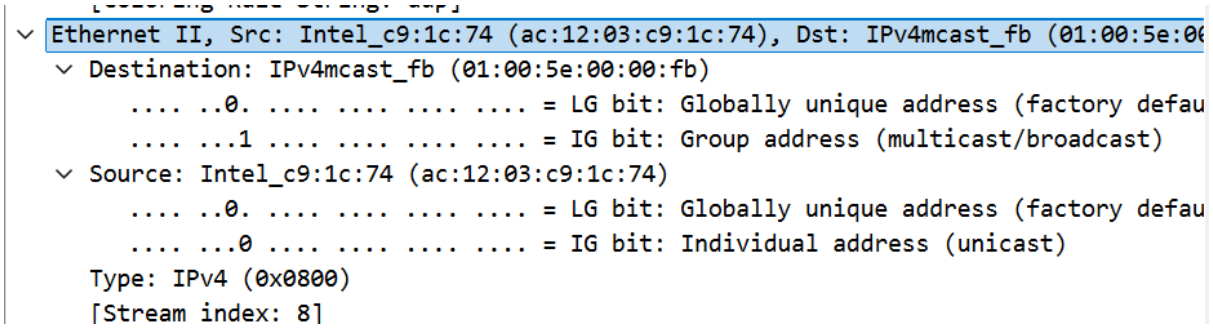
```
    Hop Limit: 1
  ▾ Source Address: fe80::946:550a:fa3a:665f
      [Address Space: Link-Local Unicast]
    › [Special-Purpose Allocation: Link-Local Unicast]
  ▾ Destination Address: ff02::fb
      [Address Space: Multicast]
    › [.... .... 0000 .... = Multicast Flags: 0x0]
      [.... .... .... 0010 = Multicast Scope: Link-Local scope (0x2)]
    [Stream index: 1]
```

Packet 4

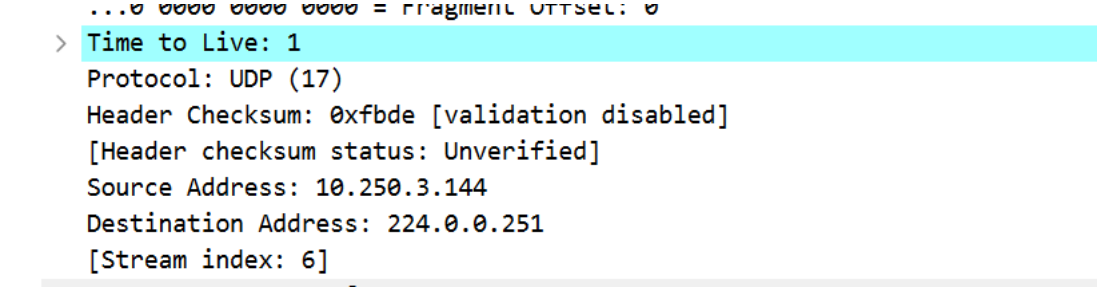
Frame number & length



Ether Type

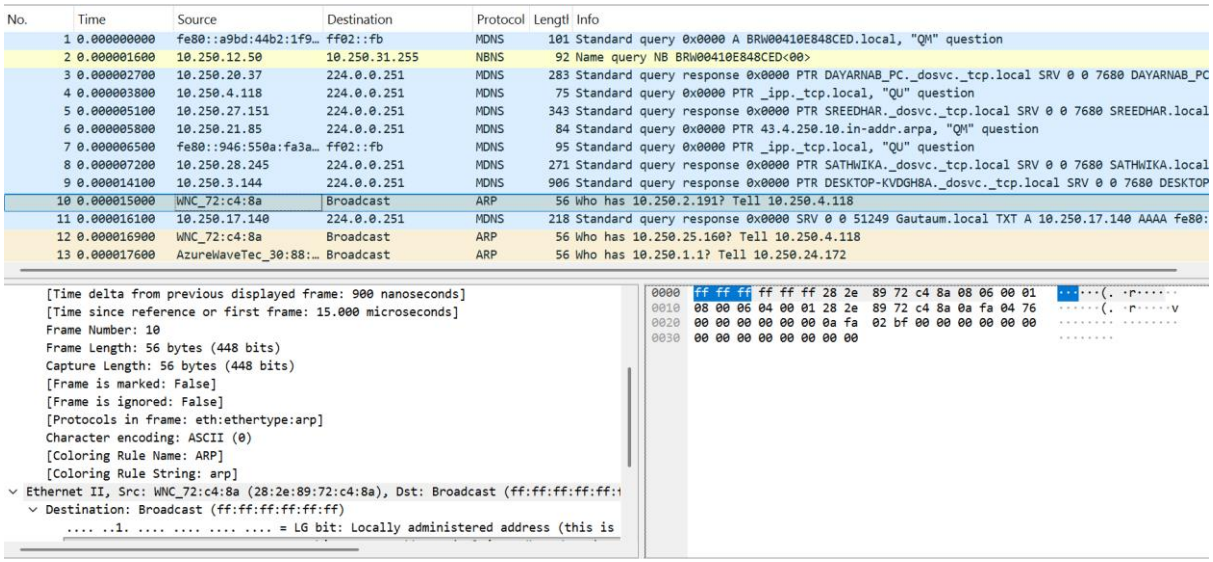


Source & destination Address



Packet 5

Frame number & length



Ether Type

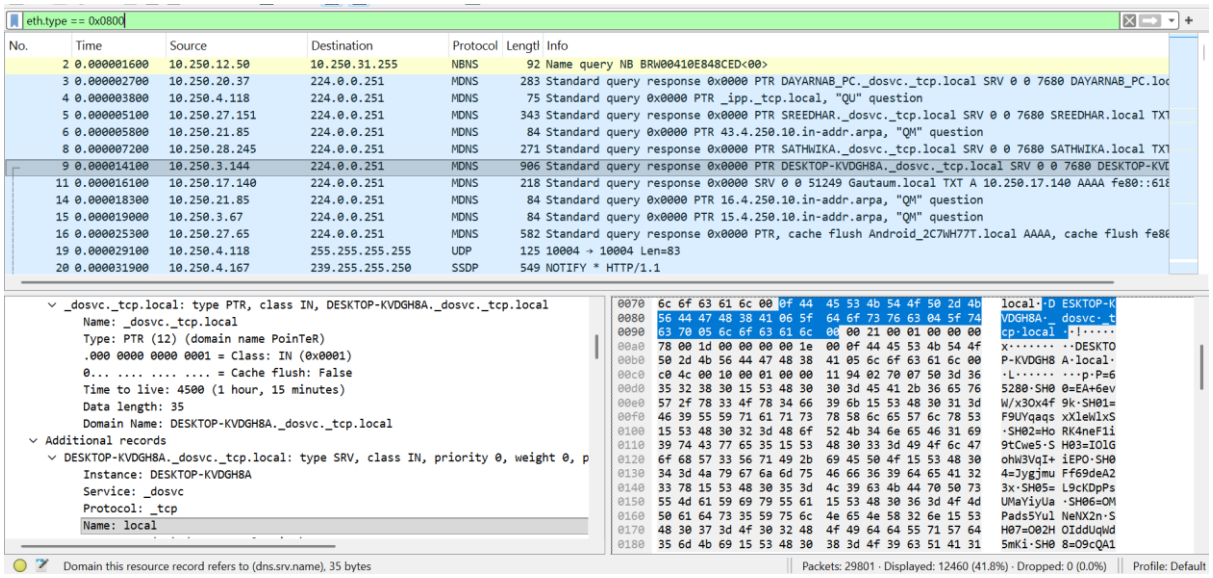
```
Ethernet II, Src: WNC_72:c4:8a (28:2e:89:72:c4:8a), Dst: Broadcast (ff:ff:ff:ff:ff:f
  Destination: Broadcast (ff:ff:ff:ff:ff:ff)
      .... ..1. .... .... .... .... = LG bit: Locally administered address (this is
      .... ...1 .... .... .... .... = IG bit: Group address (multicast/broadcast)
  Source: WNC_72:c4:8a (28:2e:89:72:c4:8a)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory defau
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 9]
```

Source & destination Address

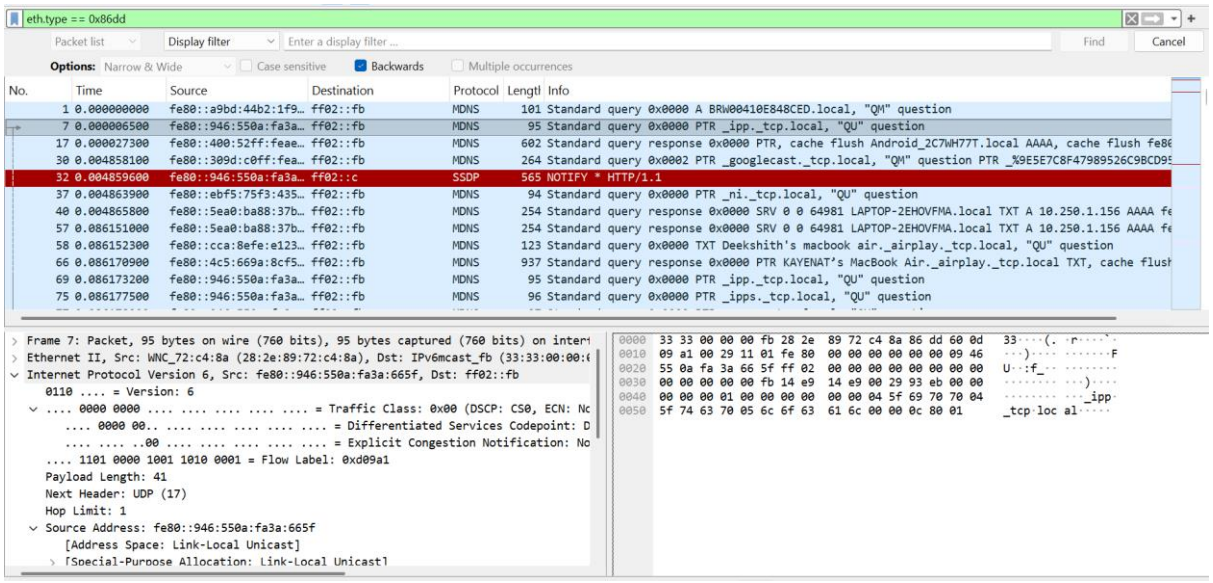
```
Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: WNC_72:c4:8a (28:2e:89:72:c4:8a)
    Sender IP address: 10.250.4.118
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 10.250.2.191
```

3. packets carrying IPv4 and IPv6 traffic

IPv4


eth.type == 0x0800
No. Time Source Destination Protocol Lengtl Info
2 0.000001600 10.250.12.50 10.250.31.255 NBNS 92 Name query NB BRW00410E848CED<00>
3 0.000002700 10.250.20.37 224.0.0.251 MDNS 283 Standard query response 0x0000 PTR DAYARNAB_PC._dosvc._tcp.local SRV 0 0 7680 DAYARNAB_PC.loc
4 0.000003800 10.250.4.118 224.0.0.251 MDNS 75 Standard query 0x0000 PTR _ipp._tcp.local, "QU" question
5 0.000005100 10.250.27.151 224.0.0.251 MDNS 343 Standard query response 0x0000 PTR SREEDHAR._dosvc._tcp.local SRV 0 0 7680 SREEDHAR.local TXT
6 0.000005800 10.250.21.85 224.0.0.251 MDNS 84 Standard query 0x0000 PTR 43.4.250.10.in-addr.arpa, "QM" question
8 0.000007200 10.250.28.245 224.0.0.251 MDNS 271 Standard query response 0x0000 PTR SATHWIKA._dosvc._tcp.local SRV 0 0 7680 SATHWIKA.local TXT
9 0.000014100 10.250.3.144 224.0.0.251 MDNS 906 Standard query response 0x0000 PTR DESKTOP-KVDGH8A._dosvc._tcp.local SRV 0 0 7680 DESKTOP-KVD
11 0.000016100 10.250.17.140 224.0.0.251 MDNS 218 Standard query response 0x0000 SRV 0 0 51249 Gautaum.local TXT A 10.250.17.140 AAAA fe80::618
14 0.000018300 10.250.21.85 224.0.0.251 MDNS 84 Standard query 0x0000 PTR 16.4.250.10.in-addr.arpa, "QM" question
15 0.000019000 10.250.3.67 224.0.0.251 MDNS 84 Standard query 0x0000 PTR 15.4.250.10.in-addr.arpa, "QM" question
16 0.000025300 10.250.27.65 224.0.0.251 MDNS 582 Standard query response 0x0000 PTR, cache flush Android_2C7WH77T.local AAAA, cache flush fe8
19 0.000029100 10.250.4.118 255.255.255.255 UDP 125 10004 → 10004 Len=83
20 0.000031900 10.250.4.167 239.255.255.250 SSDP 549 NOTIFY * HTTP/1.1
_dosvc._tcp.local: type PTR, class IN, DESKTOP-KVDGH8A._dosvc._tcp.local
Name: _dosvc._tcp.local
Type: PTR (12) (domain name PoinTeR)
.000 0000 0000 0001 = Class: IN (0x0001)
0... .... = Cache flush: False
Time to live: 4500 (1 hour, 15 minutes)
Data length: 35
Domain Name: DESKTOP-KVDGH8A._dosvc._tcp.local
Additional records
DESKTOP-KVDGH8A._dosvc._tcp.local: type SRV, class IN, priority 0, weight 0, p
Instance: DESKTOP-KVDGH8A
Service: _dosvc
Protocol: _tcp
Name: local
0070 6c 6f 63 61 6c 00 0f 44 45 53 4b 54 4f 50 2d 4b local··D ESKTOP-K
0080 56 44 47 48 38 41 06 5f 64 6f 73 76 63 04 5f 74 VDGH8A·_ dosvc·_t
0090 63 70 05 6c 6f 63 61 6c 00 00 21 00 01 00 00 00 cp·local ··!·····
00a0 78 00 1d 00 00 00 00 1e 00 0f 44 45 53 4b 54 4f x····· ··DESKTO
00b0 50 2d 4b 56 44 47 48 38 41 05 6c 6f 63 61 6c 00 P-KVDGH8 A·local·
00c0 c0 4c 00 10 00 01 00 00 11 94 02 70 07 50 3d 36 ·L····· ···p·P=6
00d0 35 32 38 30 15 53 48 30 30 3d 45 41 2b 36 65 76 5280·SH0 0=EA+6ev
00e0 57 2f 78 33 4f 78 34 66 39 6b 15 53 48 30 31 3d W/x3Ox4f 9k·SH01=
00f0 46 39 55 59 71 61 71 73 78 58 6c 65 57 6c 78 53 F9UYqaqs xXleWlxS
0100 15 53 48 30 32 3d 48 6f 52 4b 34 6e 65 46 31 69 ·SH02=Ho RK4neF1i
0110 39 74 43 77 65 35 15 53 48 30 33 3d 49 4f 6c 47 9tCwe5·S H03=IOlG
0120 6f 68 57 33 56 71 49 2b 69 45 50 4f 15 53 48 30 ohW3VqI+ iEPO·SH0
0130 34 3d 4a 79 67 6a 6d 75 46 66 36 39 64 65 41 32 4=Jygjmu Ff69deA2
0140 33 78 15 53 48 30 35 3d 4c 39 63 4b 44 70 50 73 3x·SH05= L9cKDpPs
0150 55 4d 61 59 69 79 55 61 15 53 48 30 36 3d 4f 4d UMaYiyUa ·SH06=OM
0160 50 61 64 73 35 59 75 6c 4e 65 4e 58 32 6e 15 53 Pads5Yul NeNX2n·S
0170 48 30 37 3d 4f 30 32 48 4f 49 64 64 55 71 57 64 H07=O02H OIddUqWd
0180 35 6d 4b 69 15 53 48 30 38 3d 4f 39 63 51 41 31 5mKi·SH0 8=O9cQA1
Domain this resource record refers to (dns.srv.name), 35 bytes
Packets: 29801 · Displayed: 12460 (41.8%) · Dropped: 0 (0.0%)
Profile: Default

IPv6


eth.type == 0x86dd
Packet list Display filter Enter a display filter ... Find Cancel
Options: Narrow & Wide Case sensitive Backwards Multiple occurrences
No. Time Source Destination Protocol Lengtl Info
1 0.000000000 fe80::a9bd:44b2:1f9… ff02::fb MDNS 101 Standard query 0x0000 A BRW00410E848CED.local, "QM" question
7 0.000006500 fe80::946:550a:fa3a… ff02::fb MDNS 95 Standard query 0x0000 PTR _ipp._tcp.local, "QU" question
17 0.000027300 fe80::400:52ff:feae… ff02::fb MDNS 602 Standard query response 0x0000 PTR, cache flush Android_2C7WH77T.local AAAA, cache flush fe8
30 0.004858100 fe80::309d:c0ff:fea… ff02::fb MDNS 264 Standard query 0x0002 PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD9
32 0.004859600 fe80::946:550a:fa3a… ff02::c SSDP 565 NOTIFY * HTTP/1.1
37 0.004863900 fe80::ebf5:75f3:435… ff02::fb MDNS 94 Standard query 0x0000 PTR _ni._tcp.local, "QU" question
40 0.004865800 fe80::5ea0:ba88:37b… ff02::fb MDNS 254 Standard query response 0x0000 SRV 0 0 64981 LAPTOP-2EHOVFMA.local TXT A 10.250.1.156 AAAA fe
57 0.086151000 fe80::5ea0:ba88:37b… ff02::fb MDNS 254 Standard query response 0x0000 SRV 0 0 64981 LAPTOP-2EHOVFMA.local TXT A 10.250.1.156 AAAA fe
58 0.086152300 fe80::cca:8efe:e123… ff02::fb MDNS 123 Standard query 0x0000 TXT Deekshith's macbook air._airplay._tcp.local, "QU" question
66 0.086170900 fe80::4c5:669a:8cf5… ff02::fb MDNS 937 Standard query response 0x0000 PTR KAYENAT's MacBook Air._airplay._tcp.local TXT, cache flush
69 0.086173200 fe80::946:550a:fa3a… ff02::fb MDNS 95 Standard query 0x0000 PTR _ipp._tcp.local, "QU" question
75 0.086177500 fe80::946:550a:fa3a… ff02::fb MDNS 96 Standard query 0x0000 PTR _ipps._tcp.local, "QU" question
Frame 7: Packet, 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on inter
Ethernet II, Src: WNC_72:c4:8a (28:2e:89:72:c4:8a), Dst: IPv6mcast_fb (33:33:00:00:
Internet Protocol Version 6, Src: fe80::946:550a:fa3a:665f, Dst: ff02::fb
0110 .... = Version: 6
.... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: No
.... 0000 00.. .... = Differentiated Services Codepoint: D
....00 .... = Explicit Congestion Notification: No
.... 1101 0000 1001 1010 0001 = Flow Label: 0xd09a1
Payload Length: 41
Next Header: UDP (17)
Hop Limit: 1
Source Address: fe80::946:550a:fa3a:665f
[Address Space: Link-Local Unicast]
[Special-Purpose Allocation: Link-Local Unicast]
0000 33 33 00 00 00 fb 28 2e 89 72 c4 8a 86 dd 60 0d 33····(. ·r····`·
0010 09 a1 00 29 11 01 fe 80 00 00 00 00 00 00 09 46 ···)···· ·······F
0020 55 0a fa 3a 66 5f ff 02 00 00 00 00 00 00 00 00 U··:f_·· ········
0030 00 00 00 00 00 fb 14 e9 14 e9 00 29 93 eb 00 00 ········ ···)····
0040 00 00 00 01 00 00 00 00 00 00 04 5f 69 70 70 04 ········ ···_ipp·
0050 5f 74 63 70 05 6c 6f 63 61 6c 00 00 0c 80 01 _tcp·loc al·····